Preprocessing and Input Template

Each applicant record was converted into a single text string combining structured and unstructured fields:

Program: {program}
Term: {term}
Degree: {degree}
Citizenship: {US/International}
GPA: {GPA, or "Unknown" if missing}
GRE Quant: {GRE, or "Unknown" if missing}
GRE Verbal: {GRE V, or "Unknown" if missing}
GRE AW: {GRE AW, or "Unknown" if missing}
Comments: {comments}

Missing numeric fields were represented as the literal string "Unknown" rather than dropped, so the model could still learn from the surrounding structure even when scores weren't reported. Text fields (program, comments, term, degree, citizenship) with missing values were also filled with "Unknown" rather than removed. Records were filtered to only Accepted/Rejected outcomes before training and deduplicated by URL, since Waitlisted and Interview statuses don't map cleanly onto a binary admit/reject decision.

Tokenizer Choice

DistilBERT's own WordPiece tokenizer (distilbert-base-uncased) was used rather than a custom tokenizer, since fine-tuning requires the input tokenization to exactly match what the pretrained weights were trained on. Building a custom tokenizer would have broken the alignment between input tokens and the pretrained embeddings, discarding the main benefit of using a pretrained model in the first place. A max sequence length of 256 was chosen as long enough to hold the university/program/GPA/GRE fields plus a meaningful chunk of the comments field, without excessive padding on shorter entries.

Evaluation Summary

Test set size: 5,892 examples
Test accuracy: 79.09%
Test F1 score: 0.753
Precision (Rejected): 0.79
Recall (Rejected): 0.85
Precision (Accepted): 0.80
Recall (Accepted): 0.71
Majority-class baseline: 55.36%

Model Bias and Performance Interpretation

At 79.09 percent test accuracy on a binary classification task, the model performs well above the 50 percent random-guessing baseline, and above the majority-class baseline of 55.36 percent (the share of the test set that is Rejected, the more common of the two labels). This confirms the model is learning real signal from the input text, not just exploiting class imbalance.

Examining precision and recall separately by class reveals a meaningful asymmetry: the model correctly identifies 85 percent of true Rejected cases (recall) but only 71 percent of true Accepted cases, while precision is roughly balanced between classes (79 percent for Rejected, 80 percent for Accepted). This means the model is biased toward predicting Rejected: when it's wrong, it's more likely to incorrectly reject an applicant who was actually accepted (756 such cases) than to incorrectly accept one who was actually rejected (476 such cases). A likely explanation is that GradCafe's Accepted posts are more heterogeneous in tone and content (celebratory, detailed, sometimes off-topic), while Rejected posts may share more consistent, learnable patterns.

Compared to the two-layer network from Module 12 (69 percent accuracy), this represents a roughly 10-point improvement in accuracy and is consistent with an F1 gain as well (0.753 here). This is a meaningful, sustained improvement rather than noise, and is consistent with the model having access to free-text signal (comments, program names) that the numeric-only network never saw, as discussed in the comparison section above. That said, the class imbalance in errors is a real limitation worth being upfront about: the model is more trustworthy as a signal for likely rejection than as a signal for likely acceptance.

Comparison to the 2 layer neural network

The DistilBERT model can use raw text (program names, comments), not just six numeric features like the Module 12 network. That extra signal helped as the accuracy went from about 69 percent to 79 percent, with an F1 of 0.753. Comments and program text do seem to help, especially since GRE is missing for most applicants, so the model leans on text and GPA more than test scores. The pretrained model is more flexible since it picks up on phrasing and context the numeric-only model never saw, but it's also more fragile in practice: bigger dependencies, needs a GPU to train in reasonable time, and harder to inspect than a simple two-layer network. The accuracy jump makes the added complexity worth it, though the gain is incremental rather than huge, so there's probably a ceiling on this dataset no matter which model you use.

Limitations and Ethics

Fine-tuning a pretrained model instead of training from scratch means it already understands English well, so it needs way less data and time to learn this specific task. Combining text and structured features make sense because real admissions aren't purely numeric either; comments about research experience often carry real signal that GPA and GRE alone miss. That said, self reported Grad Cafe data has clear bias. People who post here are self-selected and often more anxious or active online than typical applicants, and outcomes are self-reported with zero verification. Popular programs are probably overrepresented too. This model can be misleading because it only sees what people chose to type into a public forum, not the actual factors admissions committees use, like letters of recommendation or interview performance. Stuff like applicant pool size or funding availability that year is completely missing from the dataset. A model like this should never be used for real decisions. It's trained on an unverified, biased sample, and even 79 percent accuracy means it's wrong roughly one in five times, with no way for someone to know if they're in that group. On a public page, uncertainty needs to be handled honestly, showing a confidence score plus a clear disclaimer, not presenting the output like it means anything official. Still, a project like this is genuinely useful as a learning exercise. It walks through the full pipeline of turning messy text and numeric data into a fine-tuned model deployed behind a real web form, which is basically what real ML engineering work looks like. The value is in building and understanding that pipeline, not in the predictions being trustworthy, and that's worth being upfront about to anyone using the page.